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Smart Agriculture: Cucurbit Leaf Disease Detection Using Swin Transformer

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***Abstract*—Cucurbit vegetables such as cucumber, watermelon, bottle gourd, and pumpkin are an important part of agriculture and food security in Bangladesh and other developing countries. However, foliar diseases such as anthracnose, powdery mildew, downy mildew, and mosaic virus are significant, and the Food and Agriculture Organization (FAO) states that accurate and efficient disease surveillance systems are thus critical to food security. On the other hand, state-of-the-art deep learning techniques for plant disease classification rely on expert advice, are time-consuming, and not feasible for widespread application in resource-scarce rural settings. To address this performance shortfall, we introduce a vision transformer-based method that surpasses CNN architectures with high accuracy, class-wise robustness, and generalization capacity. Our approach evaluates two models on a multi-class plant disease benchmark: (1) A self-attention CNN that reports 88.10% total accuracy and lower test loss (0.3963), but suffers massive performance declines in significant disease classes, especially Downy Mildew (F1-score: 81.76%) and Healthy leaves (F1-score: 88.65%). (2) The Swin Transformer, an attention hierarchical vision transformer with a shifted window attention, not only achieves higher overall accuracy (92.27%) but also equally high F1 scores across all the classes, with an impressive 90.88% F1 on Downy Mildew (+9.1% improvement over CNN) and 93.36% on Healthy leaves. Training–validation dynamics and Grad-CAM visualizations also support great generalization and interpretability. Our solution can scale for application in mobile apps, drone-based scouting systems, and precision agriculture platforms that give farmers an actual-time, highly precise means of combating crop disease epidemics before they can propagate.**

Keywords—Deep learning, Swin Transformer, Vision Transformer, Image classification, CNN with self-attention model, Smart agriculture.

I. INTRODUCTION

Food security, nutrition, and rural livelihoods are greatly enhanced by cucurbit crops such as cucumber (Cucumis sativus), watermelon (Citrullus lanatus), bottle gourd (Lagenaria siceraria), and pumpkin (Cucurbita spp.). These crops are an important part of farm systems in developing countries, especially in South and Southeast Asia [1]. These crops are cultivated on a large scale due to their high market price and nutritional content, rich in vitamins, antioxidants, and dietary fiber [2]. But their yield is frequently reduced by the foliar diseases [3] such as anthracnose (Colletotrichum spp.), powdery mildew (Podosphaera xanthii), downy mildew (Pseudoperonospora cubensis), and cucumber mosaic virus (CMV), and this can cause losses in yield by 30% to over 70% if not managed [4]. Traditional disease diagnosis [5] is based on visual observation by extension agents or agronomists, which is subjective, laborious, and less accessible to remote rural farmers. The scarcity of agricultural experts [6] and delayed interventions further contribute to the spread of the disease, overuse of pesticides, and financial loss. Accurate, automated, and scalable disease detection technologies are critically needed in such a scenario. Current deep learning and computer vision breakthroughs have brought promising results to image-based plant disease detection. Convolutional Neural Networks (CNNs) such as ResNet [7], DenseNet [8], and MobileNetV2 [9] have been widely employed in various plant pathology applications with outstanding leaf disease classification performance. Nevertheless, by definition, CNNs are bound by their local receptive fields, which restrict their ability to learn long-range spatial dependencies, crucial when plant diseases exhibit weak, mixed, or heterogeneous symptoms. To counter this, Vision Transformers (ViTs) [10] have emerged as a close rival, applying self-attention mechanisms to learn long-range contextual interactions among image pixels. Swin Transformer [11] particularly introduces hierarchical design with shifted window-based self-attention to enable efficient computation and multi-scale feature learning, thus being particularly adept at handling high-resolution agricultural imagery. Recent studies by Zhichao Chen et al. [12] and Yujia Zhang et al. [13] have proven that Swin Transformers outperform CNNs in plant disease classification, but most of these efforts focus on staple crops like rice or tomato with little applicability to cucurbits, especially in South Asian contexts.

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Also, existing cucurbit disease datasets are typically small, imbalanced, or unrepresentative of true variation in the field, limiting model generalizability. In addition, none of the existing research has compared transformer models with locally sourced, field-harvested images of Bangladesh smallholder farms. Against this background, the current research makes the following contributions:

1. ***Newly Curated Dataset***: A customized collection of data from four major cucurbit vegetable leaf diseases, like cucumber, watermelon, bottle gourd, and pumpkin, as well as healthy samples, gathered from diverse agro-ecological regions of Bangladesh under natural field conditions with variations in lighting.
2. ***A Novel Application and Benchmark:*** We present one of the first rigorous empirical studies applying a Swin Transformer architecture to the challenge of multi-class cucurbit disease identification, establishing a new performance benchmark on the dataset.
3. ***Superior Robustness Through Self-Attention***: Not only does our Swin Transformer model achieve a higher overall accuracy (92.27%), but more importantly, it has great per-class consistency, eliminating key failure modes of the CNN model, most notably a 9.1% F1-score improvement on the challenging Healthy leaf class.

The remainder of this paper is organized as follows: Section 2 draws attention to related work; Section 3 discusses the novelty of this method; Section 4, discusses the methodology; Section 5, contains experimental details to assess the efficacy of the proposed model; Section 6 discusses the Conclusion.

II. Related Work

A. Deep Learning-Based Approaches in Plant Disease Detection

Early research focused on Convolutional Neural Networks (CNNs), which achieved strong performance in several datasets. For instance, Hammad Saleem M et al. [14] evaluate deep learning (DL) models (SSD, Faster R-CNN, RFCN) for classifying and localizing 26 sick and 12 healthy leaf classes in the PlantVillage dataset. SSD using Adam optimizer delivers the best performance (73.07% mAP), as compared to other architectures and optimizers (SGD, RMSProp). Shin J et al. [15] compares six CNNs (e.g., ResNet-50, AlexNet, variants of SqueezeNet) for PM detection on strawberry leaves based on 1,450 field images (augmented to 11,600). ResNet-50 achieved the best accuracy (98.11%), AlexNet achieved the least time (40.73s), and SqueezeNet-MOD2 achieved balanced accuracy (92.61%) with low memory usage, outperforming non-DL methods (e.g., SVM: 85.44%). Paper [16] proposes two deep learning models for classification of plant diseases: (1) a transfer learning DenseNet-based model (98% accuracy, 7M params) and (2) a light CNN for IoT use (70–90% accuracy, variable input sizes). An HDL model [17] that uses EfNet for feature extraction and ML classifiers (e.g., SGB, RF) for the identification of tomato early blight. Optimized with Optuna, the EfNet-B3-SGB model achieved 100% on the IARI-TomEBD dataset but faced accuracy sacrifices (up to 92.73% on PlantVillage) against computational time (35s to 33,724s). Furthermore, the work [18] presents PDDD-PreTrain, a collection of domain-specialized pre-trained models for plant disease diagnosis, filling the gap of generic models (e.g., ImageNet) by capitalizing on a 400,000+ image dataset (PDDD) spanning 120 diseases on 40 plant species. The hybrid ImageNet+PDDD model reached 95.58% accuracy, outperforming baselines in classification, detection (35.30% mAP), and segmentation.

B. Plant Disease Classification: Advances in CNNs and Vision Transformers

Jindal V et al. [19] propose a federated CNN-based approach for the detection of five prominent cucurbit leaf diseases (powdery mildew, mosaic virus, etc.) to address data decentralization and privacy concerns in Indian agriculture. Performance metrics (precision, recall, F1-score) were all above 90% and were stable across different cultivation conditions. Paper [20] proposes PDLC-ViT, a Vision Transformer-based model with localization and classification for plant disease detection. It differs from CNNs in applying co-scale and co-attention mechanisms in a multi-task model to capture global dependency with 99.97% accuracy and improved segmentation (IoU: 0.9973). Tested on PlantVillage and PlantDoc datasets, it outperforms existing models like MSCVT and MTLSegFormer. This work [21] pushes the frontiers of plant disease detection with a hybrid deep learning model based on multispectral imaging, fusing CNNs and Vision Transformers (ViTs) for higher accuracy. A carefully curated, balanced dataset (2,652 images over six spectral filters) identified ViT-B16 as the best-performing model, with 83.3% accuracy and outperforming with K720 filter data. The research [22] proposes using a light Vision Transformer (ViT) model for automatic plant disease classification compared to CNNs and hybrid models across three datasets (wheat, rice, and PlantVillage). ViT-based models achieved the maximum accuracy (up to F1-scores 0.9877) with lower computational load (0.053 GFlops as opposed to 0.794 for EfficientNet) but at reduced rates of prediction due to attention mechanisms. Hybrid CNN-ViT models balanced accuracy and speed, outperforming the base CNN without sacrificing ViT's latency. Paper [23] employs Vision Transformers (ViTs) for the classification of plant diseases, addressing global crop losses (20-40% annually, $220B economic loss) caused by homologous symptom diseases that are challenging for humans to detect. Employing a pre-trained ImageNet-21k model and data augmented (random crop, flipping, color jitter) and then fine-tuned with transfer learning, the paper utilizes the PlantVillage dataset (54,306 images, 38 classes of crop diseases). The model achieved excellent performance (99.94% test accuracy, 1.00 F1-score in validation).

III. Rationale and Novelty of this Method

While CNNs have been the top plant disease classifier algorithm, their convolutional inductive bias and local feature extraction bias are a liability for classifying cucurbits. Cucurbit diseases tend to be low-contrast, irregularly shaped lesions on

large leaf surfaces [24](e.g., mosaic virus mottling pattern) or hard to separate from each other based on color and texture (e.g., mildews). Considering these points, Swin Transformer overcomes these limitations by:

- ***Capturing Non-Local Pixel Relations:*** The self-attention mechanism allows the model to capture remote regions of the leaf image and thus discern diseases that have similar or overlapping visual appearances [25].
- ***Multi-Scale Feature Learning:*** Hierarchical architecture preserves spatial information while capturing larger contexts, which makes it suitable for disease patterns that happen across different scales.
- ***Shifted Window Method:*** Through interaction outside patch limits, the model learns symptoms that spread irregularly across the leaf surface [26].

TABLE I. WHY SWIN TRANSFORMER FOR CUCURBITS?

Cucurbit Disease characteristics	***CNN Limitation***	***Swin Transformer Advantage***
Lesions spread across leaf veins	Local filters miss global context	Shifted windows capture cross-region patterns
Visually similar mildews	Struggles with fine-grained texture	Self-attention distinguishes subtle cues [25]
Mosaic patterns (virus)	Fixed receptive field	Multi-scale attention adapts to pattern size
Healthy leaf verification	High false positives	Context-aware modeling reduces noise misclassification [24].

IV. METHODOLOGY

A. Dataset

To develop a robust model, we constructed a multi-class dataset by compiling leaf images from several reliable sources. The dataset comprises five classes: mosaic virus, downy mildew, anthracnose, powdery mildew, and healthy. In Fig. 1, a sample is shown for the five classes of leaf images.

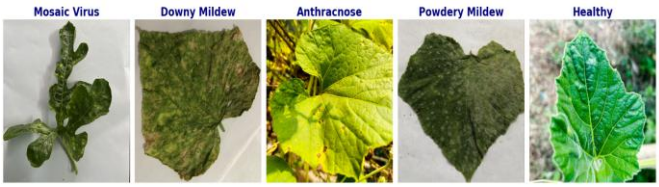

Fig. 1. Sample of Curcubit Dataset

The dataset is constructed by methodically collecting pictures from four cucurbit species commonly cultivated in Bangladesh, including cucumber (Cucumis sativus) [27], watermelon (Citrullus lanatus) [28], bottle gourd (Lagenaria siceraria), and pumpkin (Cucurbita spp) [29]. This provided extensive coverage of the cucurbit family and enhanced the model's generalization to different varieties of crops.

B. Data Pre-processing

1) Dataset Acquisition and Characteristics:

a) ***Image Source and Seasonal Variation:*** We collected our data from field images of different seasons of Bangladesh, jointly with the Regional Horticulture Research Station of Bangladesh, to obtain diverse lighting, background, high-resolution 1080*1080 pixels, and growth stages for higher robustness and generalization of the model.

b) ***Dataset Statistics:*** The dataset contains a total of 12,206 cucurbit leaf images with labels divided into five classes: Anthracnose, Downy Mildew, Healthy, Mosaic Virus, and Powdery Mildew. The precise distribution per class and the train-validation-test split are presented in Table II.

TABLE II. DATASET TRAINING, VALIDATION, AND TESTING

Dataset	Train (80%)	9767	12206 Total images
	Test (10%)	1220	
	Validation (10%)	1220	

2) Dataset Pre-processing Pipeline

Data pre-processing. Images were resized to 224×224 and were loaded from a PyTorch ImageFolder pipeline to have directory-based splits (train/val/test) as well as consistent labeling. During training, we apply realistic augmentations, RandAugment, random flip LR, small rotation (±10°), color jitter (brightness/contrast/saturation ±0.2), and random affine/zoom to simulate field variability in lighting, angle, and scale and prevent overfitting. Test and validation sets are rescaled only and are normalized to provide an unbiased estimate of performance; the test loader is built with shuffle=False to preserve ordering for metrics and confusion-matrix computation. Training used small batch sizes (BATCH_SIZE = 4) for best GPU throughput; the same random seed was always set to ensure reproducibility.

C. Baseline: Convolutional Neural Network (CNN)

We employed a strong (3×3) conv layer CNN baseline with three stacked convolution blocks, progressively increasing channels (32→64→128) and spatial down-sampling. Flattened features are fed to a two-layer classifier using ReLU, 50% dropout for regularization, and a final 5-class output layer, designed to learn hierarchical, discriminative features to classify cucurbit disease, shown in Fig. 2.

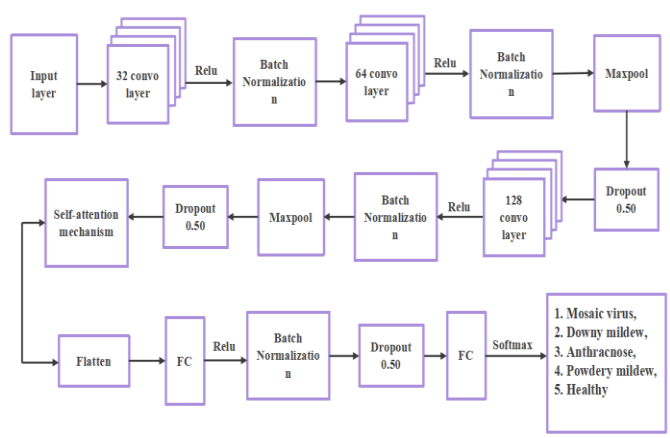

Fig. 2. CNN model framework for disease detection

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D. Proposed Model: Swin Transformer

The Swin Transformer employs shifted window self-attention to efficiently handle high-resolution crop images. As opposed to quadratic complexity for standard Vision Transformers, it processes local image windows without losing fine details (e.g., disease patches) and global context. This hierarchical processing enables accurate disease classification with linear computational complexity, as shown in Fig. 3.

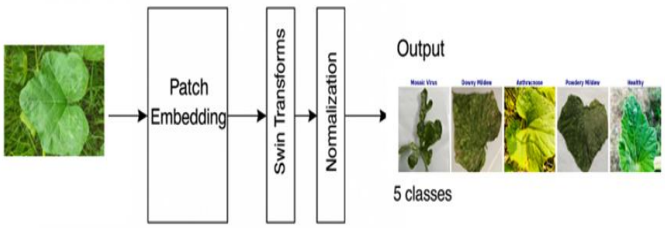

Fig. 3. Proposed model Swin Transformer framework

In every Swin block, self-attention is computed as:

- ***Step 1:*** For every patch embedding, linear projections produce Query (Q), Key (K), and Value (V) matrices.
- ***Step 2:*** Attention scores are computed inside each local window is the capacity dimension of key vectors (e.g., 32 per head). This allows every patch to focus on other patches based on similarity (QK^T), weighted by $\sqrt{d_k}$ for gradient stabilizing, and weighted by softmax for providing context-aware representations (V). Contrary to global attention calculated by standard ViTs, Swin only restricts attention to local windows with a significant reduction in computational cost.

$$Attention(Q,K,V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \quad (1)$$

- ***Step 3:*** Within shifted windows, the feature map is shifted half the window size before partitioning, enabling attention between originally non-interacting regions. Following attention, the shift is compensated for to preserve spatial coherence.

V. EXPERIMENTAL RESULT AND DISCUSSION

A. Performance Metrics

The Swin transformer model was evaluated in terms of accuracy, precision, sensitivity, specificity, and the F1 score. From this dataset, it is important to measure the accuracy of the models

- ***True positives (TP):*** Anthracnose is correctly predicted as Anthracnose.
- ***False positives (FP):*** Anthracnose (e.g., healthy or another disease) is incorrectly predicted as Anthracnose.
- ***True negatives (TN):*** Anthracnose is correctly predicted as not Anthracnose (i.e., one of the other 4 classes)
- ***False negatives (FN):*** Anthracnose is incorrectly predicted as not Anthracnose (classified as healthy or another disease).

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (2)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (5)$$

$$\text{f1} - \text{score} = \frac{(2 * \text{precision} * \text{sensitivity})}{\text{precision} + \text{sensitivity}} \quad (6)$$

The Swin Transformer is also outperforming the CNN model by 4.17% in overall accuracy (92.27% vs. 88.10%), as shown in Table III and is higher across all precision, recall, and F1 metrics, confirming its superior robustness and generalization for fine-grained cucurbit disease identification.

TABLE III. CLASSIFICATION REPORT OF CNN & SWIN TRANSFORMER

Metric	*Swin Transformer*	*CNN*
Macro Precision	0.9228	0.8857
Micro Precision	0.9227	0.8810
Macro Recall	0.9223	0.8810
Micro Recall	0.9227	0.8810
Macro F1-Score	0.9224	0.8820
Micro F1-Score	0.9227	0.8810
Overall Accuracy	**0.9227**	**0.8810**
Peak Accuracy	**0.9449**	**0.9018**

B. Confusion Matrix of Both Models

The CNN model in Fig. 4 is greatly perplexed, particularly misclassifying 15 instances of Downy Mildew as Healthy and 16 Healthy instances as Downy Mildew, illustrating its failure to distinguish visually similar pathologies, to the direct disadvantage of its overall performance. On the contrary, the Swin Transformer has high accuracy, with 298 of 300 accurate Anthracnose identifications and significantly fewer mistakes in Downy Mildew, with minimal misclassifications overall, indicating its higher ability to learn robust, discriminative features in all disease classes.

Anthracnose	297	0	1	2	0
Downy Mildew	1	268	15	10	6
Healthy	3	16	268	0	13
Mosaic Virus	2	5	3	287	3
Powdery Mildew	7	10	10	14	246
	Anthracnose	Downy Mildew	Healthy	Mosaic Virus	Powdery Mildew

Fig. 4. Confusion Matrix of CNN model

Anthracnose	298	2	0	0	0
Downy Mildew	0	290	5	2	3
Healthy	1	3	278	0	18
Mosaic Virus	1	8	0	291	0
Powdery Mildew	0	18	10	15	244
	Anthracnose	Downy Mildew	Healthy	Mosaic Virus	Powdery Mildew

Fig. 5. Confusion Matrix of Swin Transformer

C. Loss & Accuracy Curve

The training dynamics are dramatically different: the CNN in Fig. 6 model converged quickly in 7 epochs but with a higher final validation loss (~0.3), indicating poorer generalization. In comparison, the Swin Transformer in Fig. 7 was trained over 25 epochs with lower final validation loss (~0.5) and higher accuracy, with a more stable convergence and stronger ability to learn generalizable features from data.

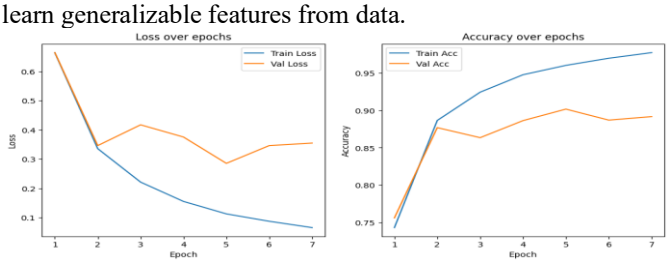

Fig. 6. Loss & accuracy curve of CNN

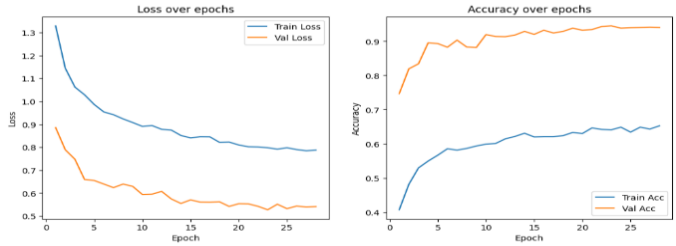

Fig. 7. Loss & accuracy curve of the swin transformer

D. Visual Interpretability by Grad-CAM

Grad-CAM heatmaps of regions of interest by disease class are presented in Fig.8. These images indicate that the Swin Transformer can identify disease-specific features, for example, mottled patterns for mosaic virus, fuzzy spots for downy mildew, necrotic lesions for anthracnose, and white powdery coatings for powdery mildew

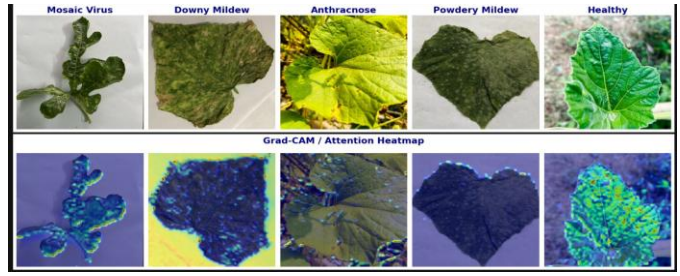

Fig. 8. Grad–CAM of the Swin Transformer Model

E. Diseases Prediction of Both Models

The Swin Transformer excels at detecting fine disease patterns, clearly outlining Anthracnose's circular lesions and Mosaic Virus's chlorotic mottling (Fig. 9). The self-attention CNN (Fig. 10), on the other hand, has difficulty with visually indistinguishable classes like Downy and Powdery Mildew and is apt to mislabel them as one another purely based on resemblance.

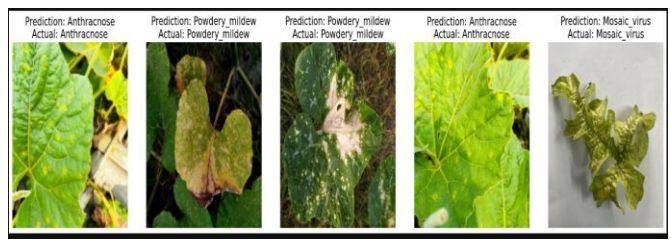

Fig. 9. Sample Diseases Predictions using Swin Transformer.

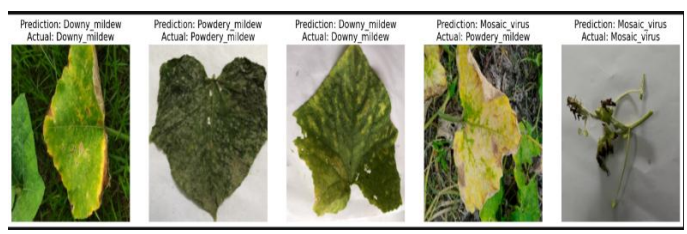

Fig. 10. Sample Diseases Predictions using CNN.

VI. Conclusion

This study presents an automated, deep learning-based approach to classifying cucurbit leaf diseases accurately with high efficiency using the Swin Transformer model. It develops a Swin Transformer-based framework for autonomous cucurbit disease detection with 92.24% accuracy on in-house field data obtained in Bangladesh. It outperformed a CNN baseline (88.1%), having higher precision, recall, and F1-score through accurate modeling of long-distance dependencies to detect fine-grained disease details, as validated by Grad-CAM visualizations. The developed model has great promise for real-world deployment. Future work will involve model lightweighting for handheld use, incorporating environmental

data to offer early warning mechanisms, and using the framework for other crops to create an integrated smart farming platform.

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